

Machine Learning for Customer Churn Prediction

Nelson David Posso Suarez

Javier Alejandro Penagos Hernandez

Nicolas Castro Rivera

1 Executive Summary

Customer Churn is a problem about customer discontinue a company's services or products. Machine learning techniques can be applied with purpose of predict if a customer has probability of be churn, this information can help the company to take decisions for not lost clients. This work search explore different machine learning techniques to compare and get the best approach to solve this problem.

2 Dataset Download and Integrity Verification

2.1 Dataset Source and Acquisition

The Telco Customer Churn dataset was obtained from the public Kaggle page published by BlastChar and identified as originating from IBM Sample Data Sets. To reproduce the setup, download the complete `WA_Fn-UseC_-Telco-Customer-Churn.csv` file from <https://www.kaggle.com/datasets/blastchar/telco-customer-churn> and place it at `data/WA_Fn-UseC_-Telco-Customer-Churn.csv`. The Kaggle page does not specify a standard open-source license; this project uses the data for academic learning and analysis and acknowledges its source and original authors.

2.2 Dataset Characteristics

The CSV is 977,501 bytes (approximately 977.5 KB) and contains 7,043 customer records and 21 columns: 20 customer and service features, plus the binary `Churn` target. It has 1,869 `Yes` and 5,174 `No` records, indicating class imbalance.

2.3 Integrity Verification

The provider does not supply an official checksum. To calculate the SHA-256 hash of the local project copy, run `pythonsrc/w1_supervised_learning/dataset_info/checksum.py` from the repository root. The script reads the CSV under `data/` and prints its checksum:

```
88BE4B93FBE0CC83421AF1C503794C97C342ECA914C1576DB7C276E61D61358A
```

This value is a local reference because Kaggle does not provide an official checksum for comparison. To generate a structured dataset report, run `pythonsrc/w1_supervised_learning/dataset_info/dataset_info.py` from the repository root. It prints JSON with the file size, row and column counts, samples per **Churn** class, each column's data type and missing-value count, and numeric feature ranges. Both scripts use the Telco CSV by default; a different file can be supplied with `--csv`.

3 EDA

Feature	Description
customerID	Unique value that identify a customer
gender	Determines whether customer is Male or Female
SeniorCitizen	Determines whether a person is citizen or not
Partner	Determines whether person has partner
Dependents	Determines whether customer has a dependent
tenure	Number of months has stayed with the company
PhoneService	Whether customer has phone service
MultipleLines	Whether the customer has multiple lines, or one, or does not have phone service
InternetService	Internet service provider (Fiber optic, DSL or no)
OnlineSecurity	Whether customer has online security, or not, or does not have internet service
OnlineBackup	Whether customer has online backup, or not, or does not have internet service
DeviceProtection	Whether customer has device protection, or not, or does not have internet service
TechSupport	Whether customer has tech support, or not, or does not have internet service
StreamingTV	Whether customer has streaming tv, or not, or does not have internet service
StreamingMovies	Whether customer has streaming movies, or not, or does not have internet service
Contract	Contract term duration (Month-to-month, One year, Two year)
PaperlessBilling	Whether customer has paperless billing or not
PaymentMethod	Payment method (Electronic check, Mailed check, Bank transfer (automatic), Credit card)
MonthlyCharges	Value paid by customer monthly
TotalCharges	Total value paid by customer
Churn	Whether the customer churned or not, stopped using the service last month.

Table 1: Dataset ³ features description

3.1 Subsection 2

Subsection example 1

3.2 Subsection 3

Subsection example 2

4 Results and Discussion

Explain the experiments you performed to demonstrate that your solution works. Summarize your unit tests in a paragraph, mentioning their philosophy, quantity, and results. Also, discuss any integration and acceptance tests to address software quality and user expectations. If possible, compare your results with other solutions using metrics. Use tables to summarize test definitions and results, and charts (such as box-plots or time series) for comparisons, depending on your project. Highlight the most important findings. Every figure and table must have a descriptive caption and be referenced in the text.

5 Conclusion

Write a couple of paragraphs to summarize your work, results, and achievements. Clearly state why your work was successful and how it helped to solve the problem. End with suggestions for future work or open questions that remain

References

- [1] Rummery, G. A., & Niranjan, M. (1994). On-line Q-learning using connectionist systems. *Technical Report*, Cambridge University Engineering Department.